

MODEL QUESTION PAPER

REVISION-2010
SUB CODE: 3086

Reg. No.....
Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY

SYNTHETIC RUBBER

Time:3 hrs

Max. marks:100

PART A

1. Answer the following questions in one or two sentences each questions carry two marks
5x2=10
1. Define Glass Transition temperature.
 2. Classify the following Polymers in to general purpose and special purpose synthetic rubbers
NBR CSM , IIR ,EPDM
 3. Write down any four main properties of Chloroprene Rubber.
 4. What are Halo Butyl rubber? Give two examples.
 5. Write any four advantages of Thermo plastic elastomer over conventional elastomer.

PART B

11. Answer any five questions each questions carry 6 marks.
5x6=30.

1. (a) What are the general characteristics of Synthetic rubber? (3)
(b) Name any three Industries producing synthetic rubber in India . (3)
2. (a) What are diene monomers? Give two examples with structure. (4)
(b) Write down the structures of Butadiene and Isoprene (2)
3. Write down the monomer preparation of Chloroprene and Acrylonitrile rubber. (6)
4. What are the characteristics of silicone polymers? (6)
5. Write three properties application of Fluorocarbon rubbers . (6)
6. Explain the production of epichlorohydrin rubbers .
7. Define Ionomers, Thermoplastic elastomers ,and Blends .

PART C

Answer one full question from each module . Each question carry 15 marks.

MODULE 1

- 11(a). Explain the production , properties and application of emulsion SBR with neat flow diagram
(8)
- (b) Explain the production , properties and application of Polybutadiene rubber with a flow diagram. (7)

OR

- IV (a). Explain the production ,properties and applications of Butyl rubber. (8)
(b). Explain the production properties and application of EPDM. (7)

MODULE 11

- V (a). Describe the effect of ACN content in the properties of nitrile rubber. (5)
(b).Describe the manufacture of silicone rubber with a neat flow diagram. (6)
(c). Write down the curing mechanism of Chloroprene rubber.

OR

- VI. (a) Explain the manufacturing of CSM. Give its characteristics. (5)
(b) Explain the curing mechanism of silicone rubber. (6)
(c) What are the monomer of Polyurethane ? Write their structure .

MODULE 111

- VII. (a). Explain the manufacturing and properties of HTPB. (6)
(b) Describe the characteristics and application of Acrylic rubber. (5)
(c) What are the important properties and applications of Poly propylene rubber.(4)

OR

- VIII (a) Explain the manufacturing and applications of CTPB.(6)
(b). Describe the characteristics and applications of Hydroxy terminated NBR.(5)
(c). What are the important application and properties of Epychlorohydrin rubbers.(4)

MODULE 1V

- IX (a). Explain the manufacture of S-B-S Thermoplastic elastomers.Also mention their application and properties. (8) (b) Describe the manufacturing of NBR/PVC blend with a typical formulation. Mention its characteristics and applications.(7)

OR

- X (a). Explain the production,properties andapplication of S-I-S thermoplastic elastomers (8)
(b) Describe the manufacture of NR-PB blend gives its application and properties.(7)

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